Moving To Edmonton Made Easy

Design Document

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Aaron A, Ryley G,
Omar M, Vlad M

WHAT?

WHY?

HOW?

INTRODUCTION

Edmonton significantly represents the multiculturalism of Canada with its diverse population; people from within or outside the city are keen on learning what a good neighborhood to live in is. It is not exactly easy to come to such a conclusion. We decided to put ourselves into the shoes of someone wanting to move into Edmonton. We know people's choices are typically limited by their budget, and they would like to live in safe neighborhoods with people who speak the same language. Our application will revolve around the qualities mentioned above, as it meets the needs of most people who want to move to Edmonton.

Our application aims to help users find suitable neighborhoods to live in without the hassle of going through multiple resources. To provide an excellent user experience, we use a methodological approach to presenting data to users through data visualization techniques.

What: Data sets

There are four datasets that we will be using in the Moving to Edmonton Made Easy dashboard. We obtained the datasets from Government open data sites such as data Edmonton and EPS' (Edmonton Police Service) website. The first dataset represents a 2016 census conducted by the city of Edmonton, where they recorded household languages and categorized the data by neighbourhood and ward attributes. The dataset contains records of 11 languages (Arabic, Cantonese, English, Filipino, French, German, Mandarin, Indigenous, Spanish, Punjabi, and Ukrainian) and a generalized column named "Others" representing minor languages. The second dataset contains value assessments of properties in Edmonton, which consist of information such as the property's value, address, ward, neighbourhood, latitude and longitude, and assessment classes. The third dataset contains a recent 90-day report of crimes in Edmonton. The dataset categorized the quantitative values by crime types, such as violent or non-violent, date of occurrence, and coordinates (location of occurrence). Lastly, we will be using the Edmonton neighbourhood map view dataset's multipolygon attribute to trace the neighbourhood in the map User Interface we will be integrating in the dashboard.

We were able to utilize the datasets together as they contain related fields such as neighbourhood and ward attributes; with these attributes, we were able to group the quantitative values of property assessments and languages spoken. We derived the provided data by providing a context for our project, the context being a target user looking for a place to move to in

Edmonton. During our derivation of the dataset, we normalized attributes such as languages by taking the overall population of a selected neighbourhood and using that to calculate the ratio of each language in that neighbourhood. We also derived the property assessment data by producing the average values of the properties in Edmonton neighbourhoods and derived the crime report dataset by calculating the rate of crime occurrence based on the day of the week. The resulting derived data will illustrate the percentage and distribution of languages, crime rate, and average property values in Edmonton neighbourhoods and wards.

What: Design Choices

The main functionality of our dashboard will be to act as a tool for the user to identify areas and neighbourhoods in Edmonton which would suit their needs. The primary focus point of the visual dashboard will be a map view of the city with visible neighbourhoods' boundaries which can make the neighbourhood selection process easy and accurate. To appeal to the user's iconic and short-term memory, the map view will allow a broad overview of the city as the welcome setting to the page. As the user hovers the mouse on the map, the corresponding neighbourhood will be highlighted, and the name will be displayed for clicking accuracy, creating a pre-attentive attribute and focusing the user's attention. The information the dashboard displays is primarily categorized into a spatial area; therefore, a map view is the best choice for delivering the data effectively. The map will be equipped with a

property value slider filter which will highlight the neighbourhoods that fall within the given range. A languages dropdown menu will serve a similar purpose in highlighting the neighbourhoods which have a high population of target language speakers.

On the left side of the map, the dashboard will have a selection and filtering menu where the user can select various attributes based on their area of interest. A neighbourhood search input box will be available for searching a particular neighbourhood and focusing the user's attention on that specific area. Within the menu, drop-down boxes will be available for selecting crime statistics, property assessment values and languages spoken to be displayed on the map, as well as on visualization graphs. As the user is focused on a neighbourhood and based on the filtering criteria they select, the area below the map is reserved for displaying graphs representative of the statistics related to that location. Bar graphs will represent these statistics to visualize quick comparisons, and line graphs to reveal trends within that neighbourhood. Finally, a dashboard section will be reserved for selecting two or more target neighbourhoods for a head-to-head comparison of their statistics. The entire menu will contain drop-down boxes and radio buttons where applicable, with all features serving as affordances for the dashboard. The simplicity of the filtering process is designed to be intuitive and very familiar to most users.

The dashboard will have a predominantly blue style with various hues throughout, used to create a bright environment that uses white space as well as contrast to highlight important areas and sections. The graphs sections will be colour highlighted with a variation of green for property

assessments, red for crimes and yellow for languages. For accessibility, a dark mode version will also be available.

Why: User Cases

Chris De Leon

age: 45
residence: Manila, Philippines
education: Computer Technology
occupation: Business Manager
marital status: Married with kids

I would like to move to a place where my family could have a better future.

Chris is a technology store manager who is currently based in the City of Manila, Philippines. He applied to immigrate to Canada with his family and has been recently approved. With this news, Chris is currently searching for Canadian cities where he and his family could relocate.

Comfort With Technology

INTERNET

SOFTWARE

MOBILE APPS

SOCIAL NETWORK

Criteria For Success:

Find an application that could help me in my research for finding a new home where my family can transition well.

Needs

- Straightforward application
- User friendly and easy to use application
- Reliable and accurate information

Wants

- Simple User interface that will not overwhelm or confuse me
- Responsive and updated data
- Dark mode

Values

- Detailed representation of information
- Accurate visualization
- Time saving

Fears

- End up in the wrong city or place in general
- Inaccurate information
- Time consuming

Veronica Suárez

age: 31
residence: Edmonton, Alberta
education: Accounting Diploma
occupation: Accounting Technician at Pathfind LTD.
marital status: Single, no kids

The only thing more important than my new SUV is my poodle, Samwise.

Veronica works a 9-5 style job with a set schedule that leaves her free on evenings and weekends to enjoy long walks with her 2 year old poodle, Samwise. She is currently renting an apartment but currently looking to find a permanent area that is relatively close to her Downtown office and within her budget. She recently made her first big purchase, a 2023 BMW X5 which she takes care of very carefully. Her hobbies include salsa dancing and cooking her mom's famous paella with the fresh ingredients found at her local Spanish market.

Comfort With Technology

INTERNET

SOFTWARE

MOBILE APPS

SOCIAL NETWORK

Criteria For Success:

Discovering an easy to use application that will show her information about different parts of Edmonton, putting her at ease about her decision to move to a new neighborhood.

Needs

- Simplicity in the search process
- Inclusiveness of safety information
- Combining location with a broad view with neighbourhood home prices

Wants

- Easy discovery of a community that potentially have Spanish grocery stores
- A good balance of safe and affordable neighbourhood

Values

- Safety
- Easy planning with no headaches
- Attention to detail

Fears

- Her new car being vandalized
- Harassment while walking her dog
- Difficulty finding the Padron peppers she grew up used to eating

Robert Smith

age: 61
residence: Edmonton, Alberta, Downtown
education: PHD, Computer Science
occupation: Retired
marital status: Single

We can hope to be fortunate—but we should strive to be wise

Robert loves spending his time's researching algorithms, he has been programming for 40 years and gained a wealth of knowledge. He is an avid volunteer at startup Edmonton, ever since the pandemic happened Robert noticed an uptick of crime in his area. Within the last 2 years, he was robbed and assaulted. Robert needs a change of scenery so he can live without fear.

Comfort With Technology

INTERNET

SOFTWARE

MOBILE APPS

SOCIAL NETWORK

Criteria For Success:

Wants to live in a safe neighbourhood where i can take my morning walks without fear.

Needs

- Find neighbourhoods with low occurrences of crime.
- Map-view of neighbourhoods so he can know if there are grocery stores nearby,
- The application should have recent data.
- Able to filter neighbourhoods by crime type

Wants

- Would like to live in a neighbourhood with German speakers.
- Wants a user accessible application for his poor eyesight
- Wants the application to be easy to use

Values

- Data privacy
- User accessible applications
- Applications that have good UI/UX

Fears

- Fears that he will be robbed or assaulted .
- Fears that he will get data that is not recent.
- Fears that the application will be hard to use

How: Prototype Overview

Figure 1 is showing the main page. On it we have a map of the city of Edmonton, with filters for assessment data, spoken languages, and crime statistics. Directly underneath the map we have tabled data for assessment values, and graphs for crime and languages. Further down the page and clearly separated from the map we have comparisons between two selected neighbourhoods, showing data for the assessment value, language and crime statistics.

Overview Figure 1.

The user can choose a specific neighbourhood through the search bar at the top of the sidebar filter options. They can also select a specific neighbourhood by utilizing the map, which will be highlighted first with all 12 wards within the city. Upon selecting an area in the map within one of the wards, the map will focus on that ward and display the neighbourhoods within it, eliminating neighbourhoods not within that ward. The selected neighbourhood will then be displayed within the search bar.

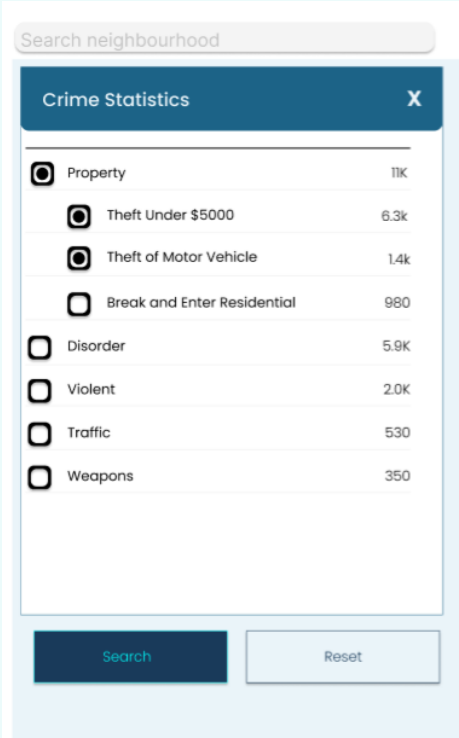

Fig 2. Filtering

The slider on top of the map will filter neighbourhoods on the map within the selected ranges, and the button beside it will allow filtering of properties within the neighbourhoods based on residential status or commercial status. Within the map we see a dropdown button that will filter neighbourhoods based on the selected language.

The sidebar on the left shows the filters for the crime statistics; property; theft under $5000, theft of motor vehicle, break and enter residential, disorder, violent, traffic, weapons. Each of these can be selected through a checkbox and will display the amount of each crime right-aligned.

Two buttons lie below the crime filter, search and reset. Search will update the map when filters are adjusted. Selecting a different language within the language dropdown, or moving the assessment sliders to new values will not be displayed until the search button is pressed. The reset button will set all values on the page back to how they were before any selections were made.

Directly below the map we see the calculated data from the filtering. The assessment data will display the average, minimum, maximum, and full list of data within a table for the average assessed value in each neighbourhood. While first on the main page, this data will be displayed related to the grouped ward data. After selecting a ward, the data will show the assessed value for those neighbourhoods. When a neighbourhood is selected, the average assessment value will only be displayed.

Fig 3. First set of visualizations

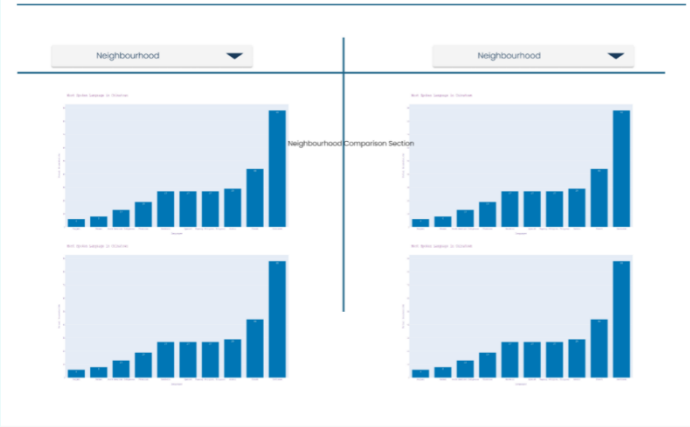

Fig 4. Comparison visualization

Beside the assessment data, there will be a crime data graph displaying all sorted crime information based on user selection, showing the wards in the city, neighbourhoods within a ward, and the neighbourhood. The language graph next to the crime graph will be displayed the same as the crime graph. Separating the map with a visually distinct line, lower on the page will be a comparison between two neighbourhoods. This comparison does not interact with the filters higher in the page. Two neighbourhoods will be selected, and graphs will display the average assessment value, crime information, and language distribution for each neighbourhood to create a visual comparison between the two neighbourhoods.

Design Justifications

We aim to give the user as many options as possible to filter the data. While also keeping the webpage simple to use and easy to understand.

Budget filter: Using a slider to select your budget range is best since the user can filter through prices quickly. Typing a range through your keyboard will be inefficient and annoying for users.

Crime filter: There are many crime types, to cause less confusion for the users, we made five main crime types, and upon pressing one, you will see subcategories of that crime type. Suppose someone is only interested in property crime. They can view a subcategory of that crime, such as “Break and Enter” and select it.

Language filter: A Combobox with language selections should be familiar to users and easy to find.

For the users to apply these filters should give the user an interactive map-view of where neighborhoods are located. This is superior to displaying a table because the user can explore the map and look for other details about the surrounding neighborhood.

At the bottom of the map, you have a comparison chart; the user being able to stack up neighborhoods against each other will lower the cognitive load of remembering the differences between two neighborhoods.

Expressiveness and Effectiveness Principles

Effectiveness principle: the most important attributes should be encoded with the most effective channels in order to be most noticeable, and then decreasingly essential attributes can be matched with less effective channels. (Munzner, 2015).

The budget slider is the most important attribute since people who are looking for homes do not have an infinite budget. The “position on a common scale” is the most effective channel in this instance because it will make our budget slider the most noticeable.

Expressiveness principle: visual encoding should express all of, and only, the information in the dataset attribute. (Munzner, 2015).

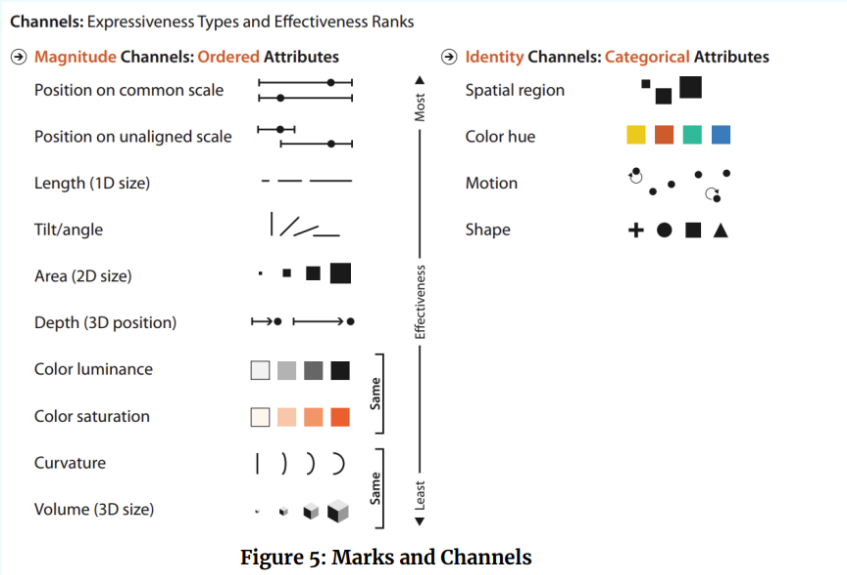

Fig 5. Marks and Channel

Since our data contains categorical data such as crime occurrence, date, and locations, we used bar charts to encode attributes. The vertical spatial position channel contains quantitative attributes such as crime count. For example, the horizontal spatial position channel will contain categorical attributes such as Assault, Robberies, and Break and Enter.

Rules of Thumb

2D Justification

Our primary focus point within the dashboard is a planar interactive map which not only attracts the users' attention but also represents a familiar mode of interaction for most people. Instead of displaying relevant data to the user in a table format, the jump from a 2D table to a 2D map is relatively straightforward. Although the map will allow for zooming into an area of interest, the nature of the planar view will maintain a 2D layout. The filtering system allows the user to select target neighbourhoods which will trigger a layer to highlight that specific neighbour for clarity. The 2D layers work well in combination and do not distort the spatial information conveyed as they are displayed in an (x,y) coordinate system. A 2D format similarly suits the crime and language charts showing trends, as the observer is not confused by distortion or complexity. When observing the bar and line charts, text information also appears on mouse hover events for clarity in cases where categories are closely related. Ultimately, a 2D representation is the safest format for user familiarity and requires one of the lowest levels of cognitive load.

Eye Beats Memory

The general focus for the dashboard is to be easy to use and intuitive; limiting the users' reliance on memory is highly important. The map information displayed will be limited to a few different colours, ensuring the user does not have to remember many categories and their corresponding

hues. Grouping relative topics close together, such as graphs, in a particular area will also limit the users' need to memorize what that specific area is meant to do. The comparison section of the dashboard will be laid out in a way which displays the neighbourhood clearly, reducing confusion as to which target area the information is referring to. Text and titles are used in a balance between an informative and a non-cluttered feel to the dashboard, meant to subtly notify the users' attention on what each section represents. Finally, all the components that make up the dashboard are chosen to make the users' task flow as straightforward as possible, with tools they would have all seen before and do not require much of a learning curve.

Overview First, Zoom and Filter, Details on Demand

Exploring information collections becomes increasingly difficult as the volume grows (Schneiderman, 1996). An interactive User interface provides the users with dynamic feedback for identifying changes by allowing interaction with the visualization. This also provides users with general information, such as the orientation or context of the visualization.

To apply this rule of thumb, Move to Edmonton Made Easy contains a dynamic mapping feature where the users will initially be provided with a map of Edmonton as an overview of the information orientation or context. The map will be interactive, so they will be able to zoom in on a specific neighborhood they choose to investigate. Filters for languages, neighbourhood, and property assessment values will also be integrated as this will narrow down specific information the users will need and, at the

same time, remove unnecessary data. The user will be provided the details based on the selected neighbourhood and/or filter inputs. The details will come in the form of visualizations such as graphs.

Responsiveness is Required

Due to the interactive nature of the Move to Edmonton Made Easy dashboard, responsivity becomes a requirement. Without responsiveness, users may find it unsatisfying or challenging to track their actions when performing general user tasks. For example, the map provided in the dashboard is clickable; if the user hovers and scrolls around the map without receiving any response from the interface, the user may think that the Edmonton neighbourhoods are not interactive at all. This could lead to them missing the opportunity to have a convenient method of filtering through the map instead of having to type or search for the neighbourhood they are interested in.

To address this concept, the map component, along with buttons in the dashboard, will provide users with the necessary response based on their actions. The neighbourhood plotted on the Edmonton map will be highlighted when a user hovers over it. Popup messages will also be implemented to inform the user when the user gives an invalid input.

Get it Right in Black and White

The aspects of the page should be visible even when the website has no colour. Each of the sections we are focusing on such as the map, the filters, graphs and numerical data, and comparative data between the neighbourhoods are still visible while lacking hue.

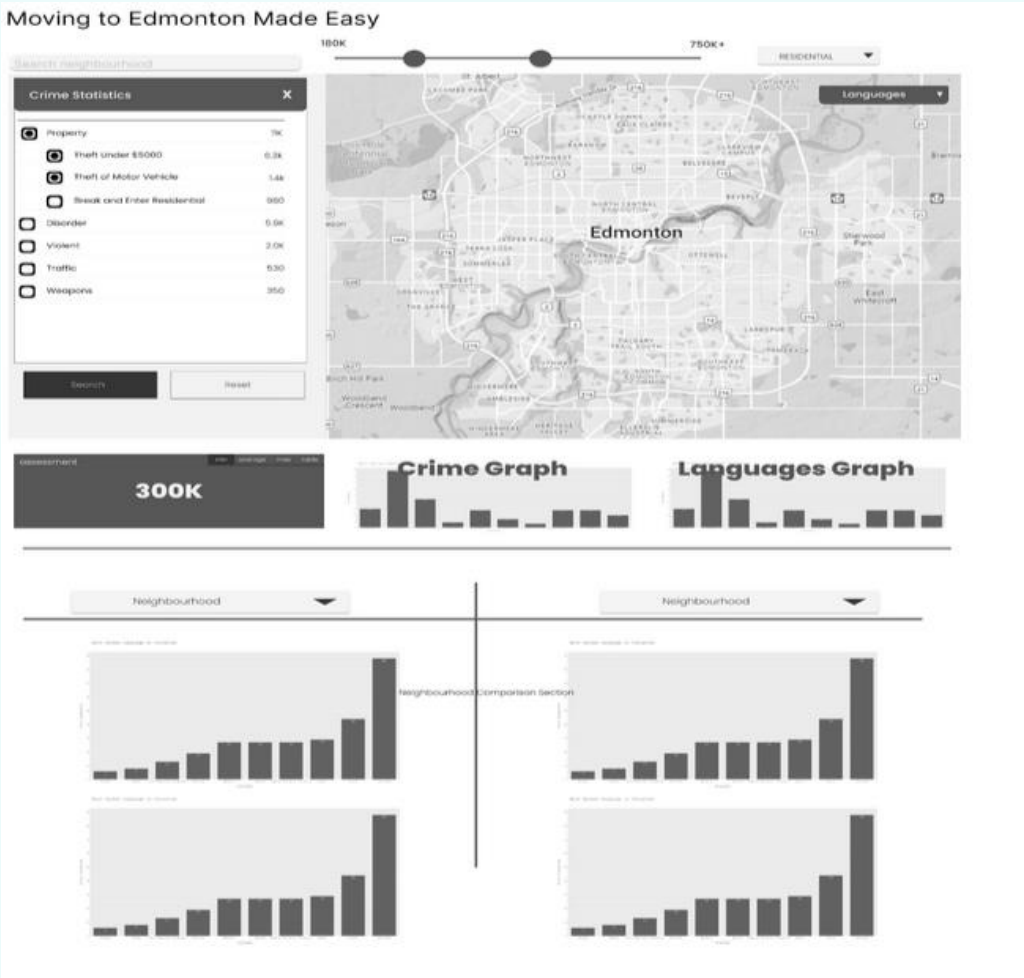

Fig 6. Grayscale Dashboard

Function First, Form Next

Ensuring that the function of the design is accurately conveyed is considered more important than its initial look. Changing the look of the design is much easier compared to making the function effective.
Before beginning our prototype, we considered the features that would be necessary for the user to interact with. Features utilized within such as filtering, searching, and multiple forms of visualization to easily represent differences in datasets and create locational grouping.

We curated the design to focus on the averaged neighbourhood data instead of individual properties, which is reflected in all of the features. The map allows visual browsing for finding a specific neighbourhood, while the search bar allows immediate results. The filters are specific in their scope of focus on neighbourhoods. Below the map the statistical information for the three datasets has the primary purpose of informing the user on the selected neighbourhood.

The comparison between two neighbourhoods, separate from the filtering, provides the user with further visual representation and informing the user between two choices.

Style Tile

The style tile provides a theme for the dashboard and the styles used in creating it. The colors are chosen for the specific reason of reminding the user of the Edmonton city website and keeping a common theme.

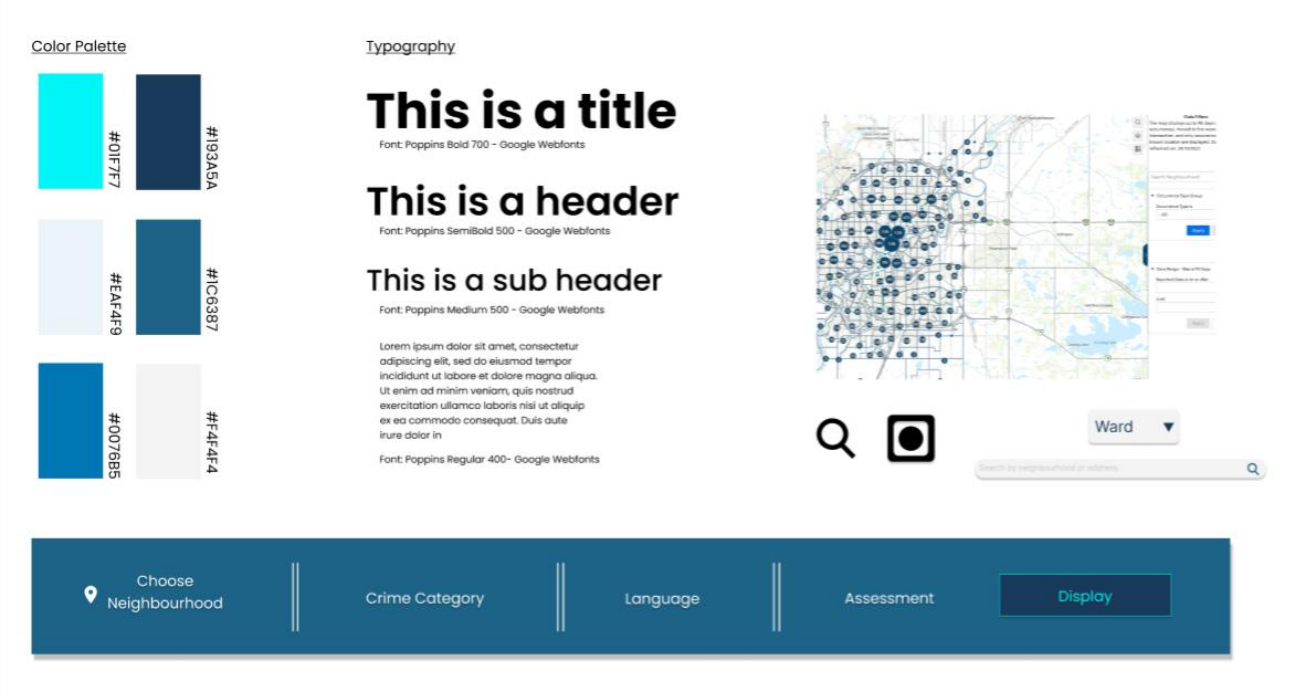

Fig 6. Style Tile

References and Resources

Personas - https://personagenerator.com

Wireframes - https://www.figma.com

Munzner, T. (2015). Visualization Analysis and Design, CRC Press.

Schneiderman, B (1996). The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations, IEEE

Datasets:

2016 Census - Dwelling Unit by Language (Neighbbourhood/Ward)
https://data.edmonton.ca/Census/2016-Census-Dwelling-Unit-by-Language-Neighbourhoo/jsc3-gmwb

City of Edmonton - Neighbourhood (Map View)
https://data.edmonton.ca/City-Administration/City-of-Edmonton-Neighbourhoods-Map-View-/wps4-8auk

Property Assessment Data (Current Calendar Year)
https://data.edmonton.ca/City-Administration/Property-Assessment-Data-Current-Calendar-Year-/q7d6-ambg

EPS : Community Safety Data Portal
https://communitysafetydataportal.edmontonpolice.ca/pages/trends